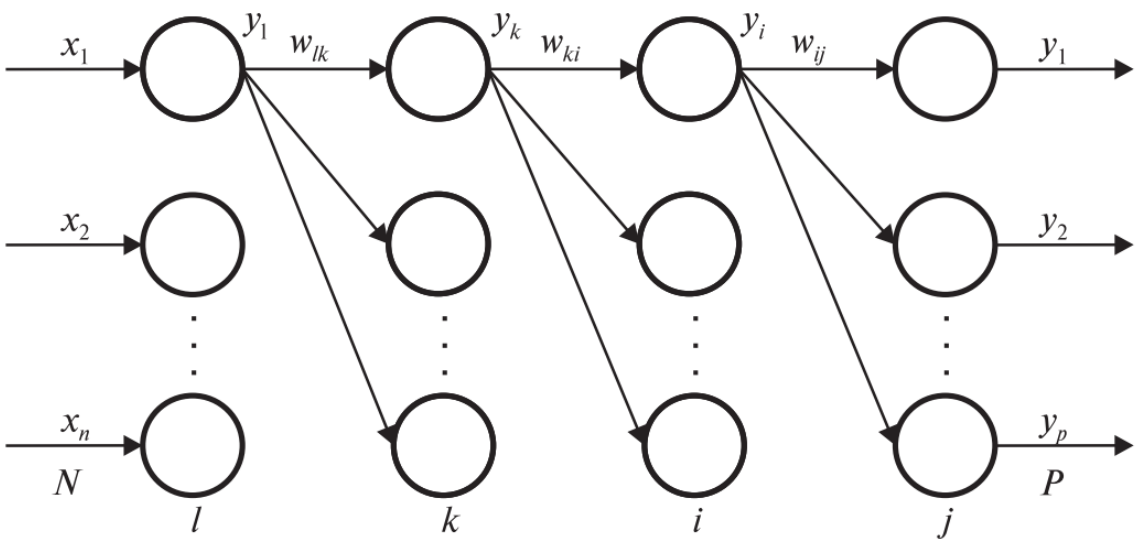
$P = \frac{t}{p}$

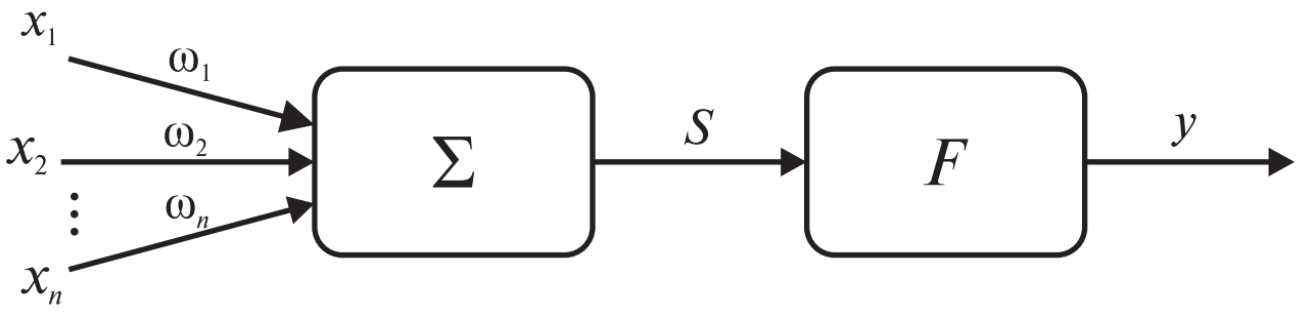
t

p

$t = 36.84 - 048T$

T°





$$S = w_1 \cdot x_1 + w_2 \cdot x_2 + \cdots + w_n \cdot x_n + b$$

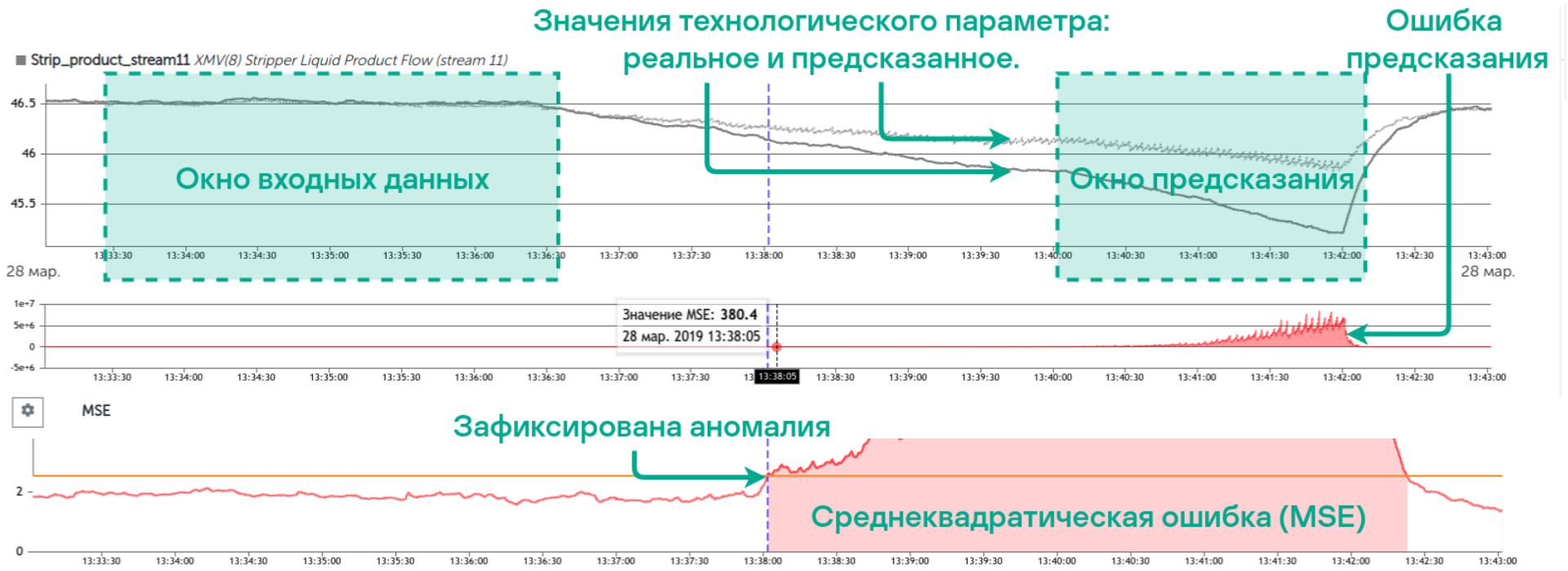
S
x_n
w_n
b

$$y = F\left(S\right)$$

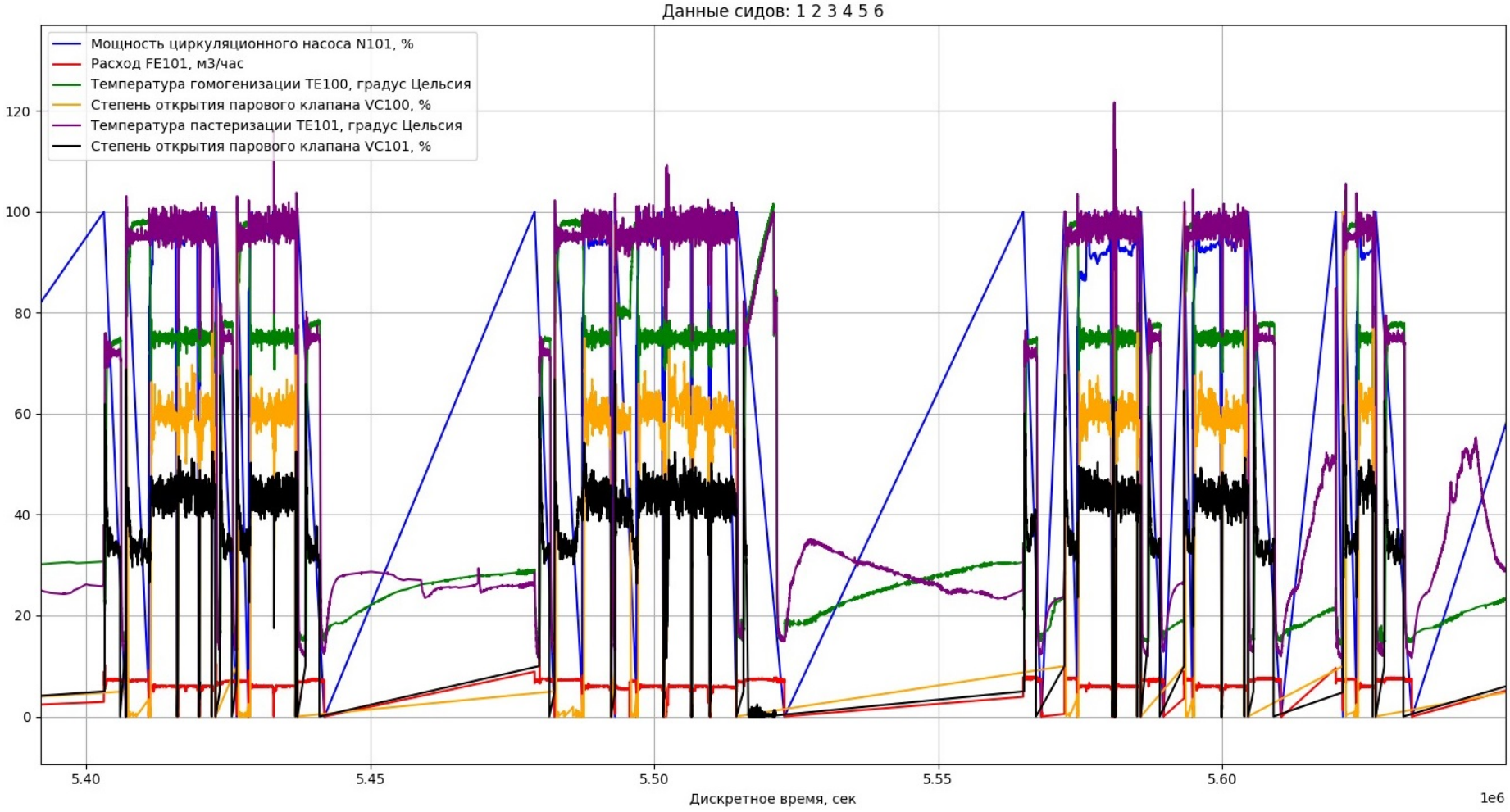
$F(...)$
y

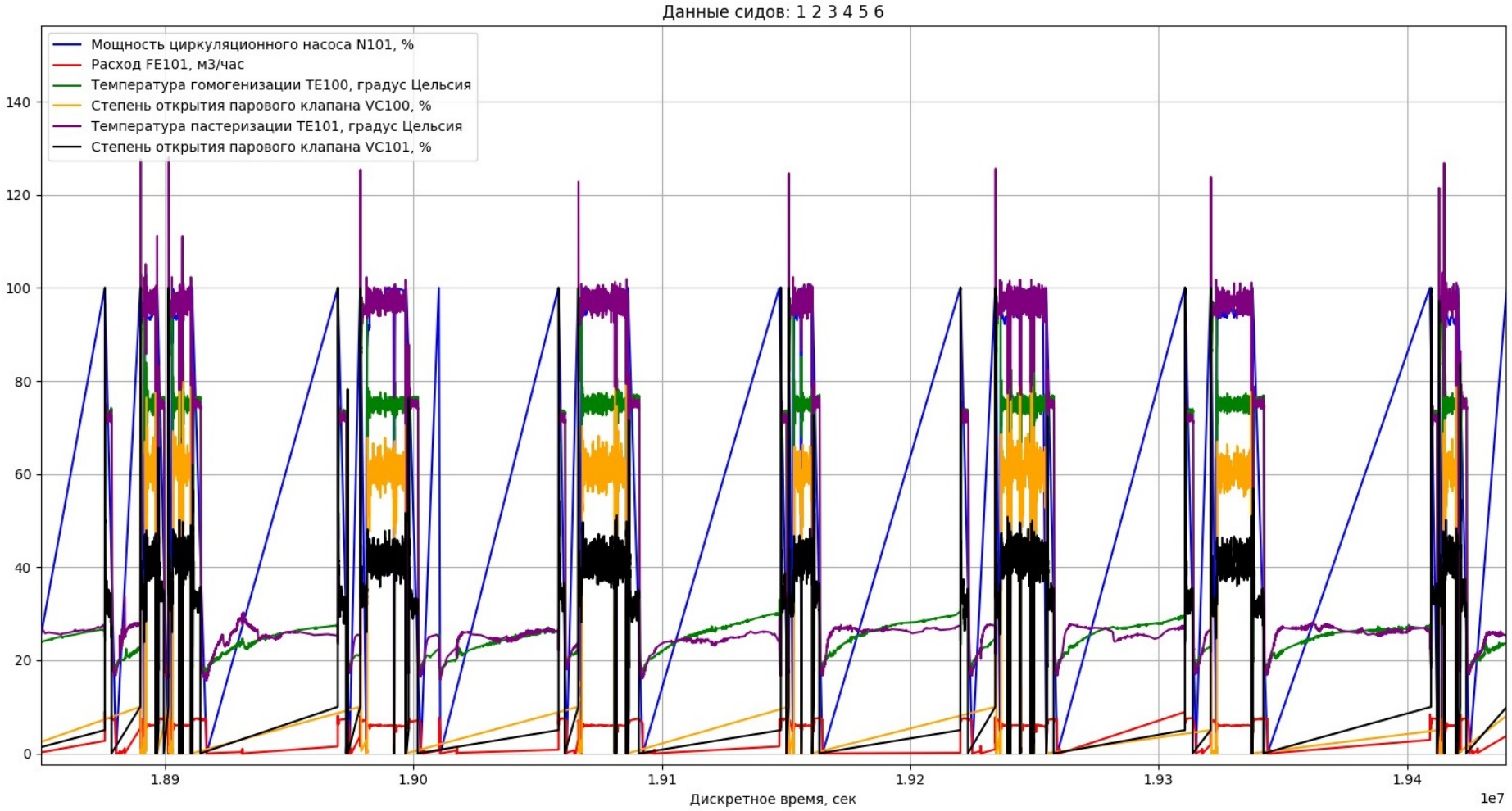
$$\{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}\{(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)\}\{x_1, x_2, \ldots, x_n\}$$

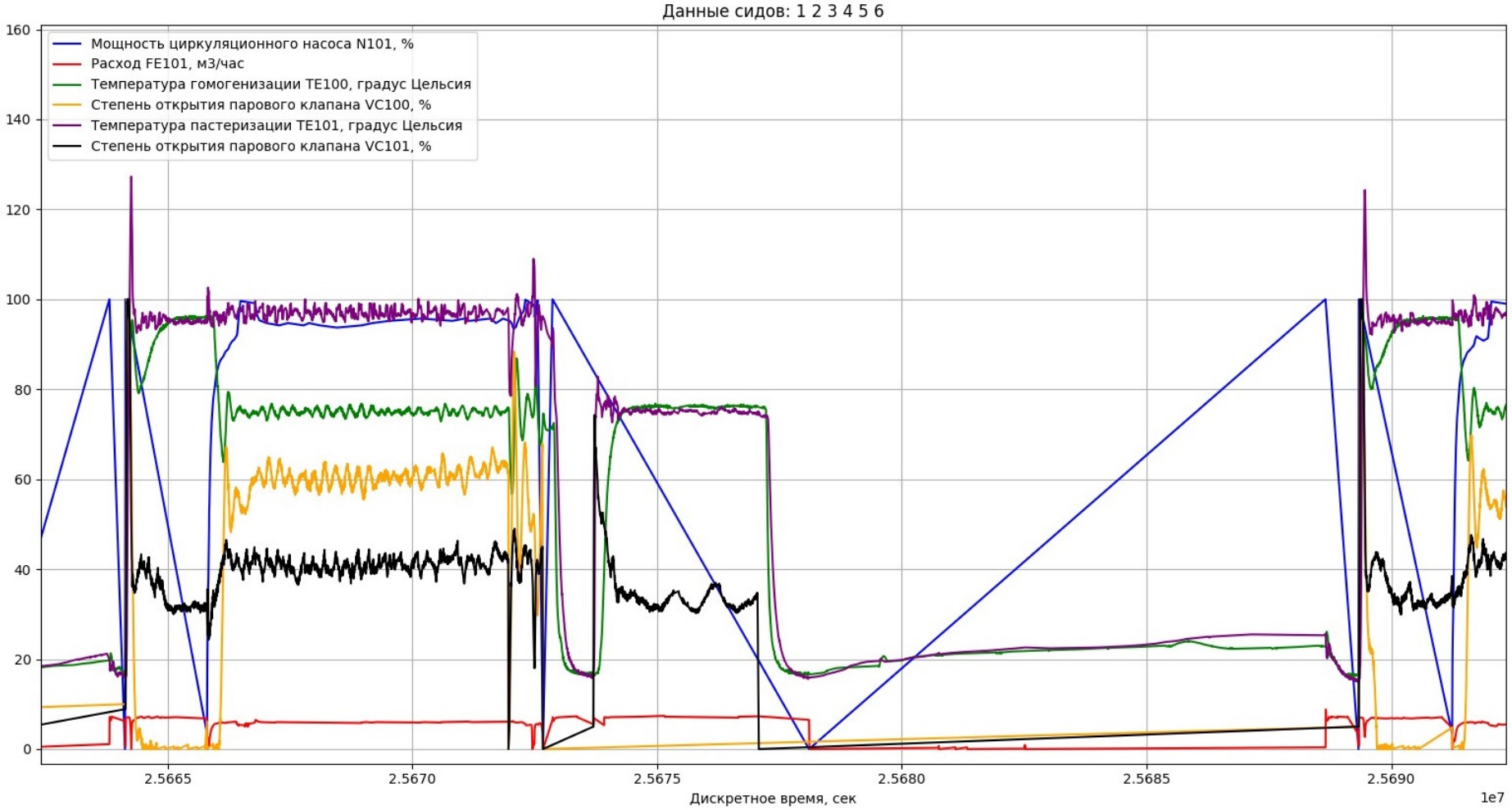
$$\{x_1, x_2, \ldots, x_n\}$$

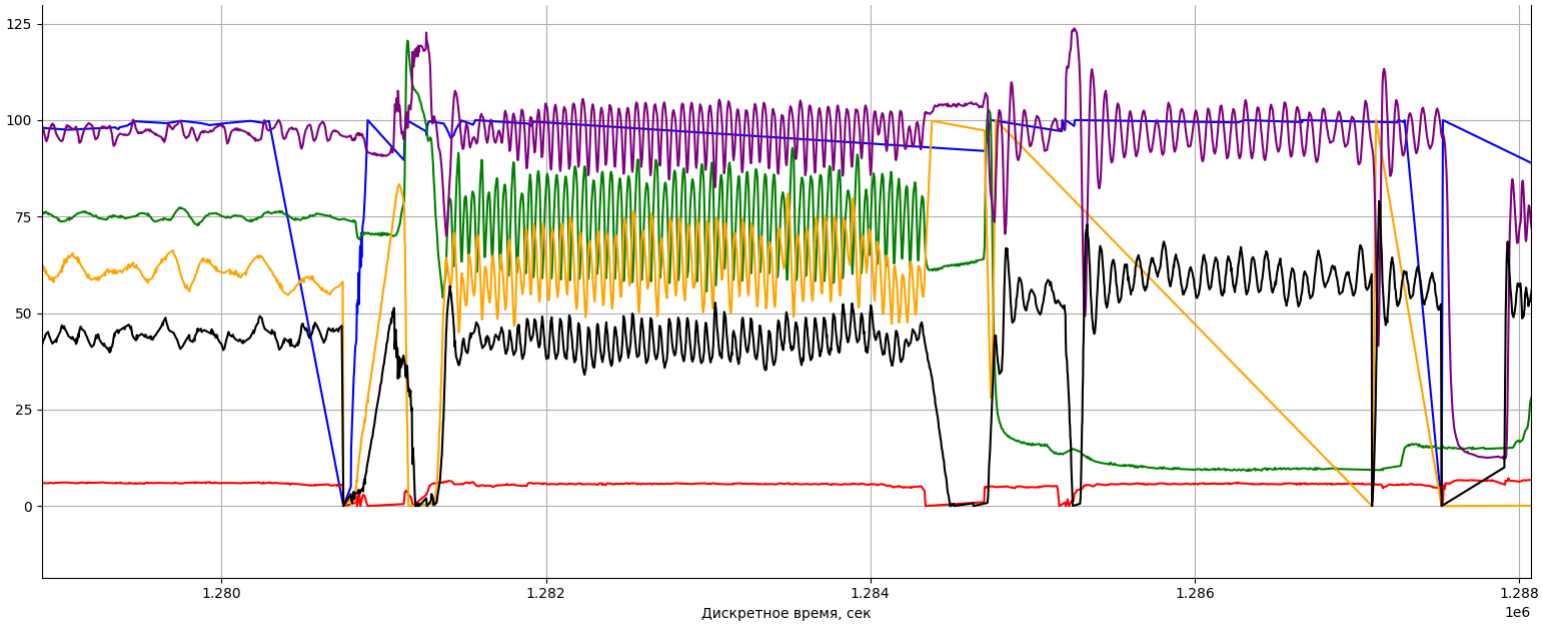

Значения технологического параметра:
реальное и предсказанное.
Ошибка
предсказания
Strip_product_stream11 XMV(8) Stripper Liquid Product Flow (stream 11)
Окно входных данных
Окно предсказания
28 мар.
28 мар.
Значение MSE: 380.4
28 мар. 2019 13:38:05
MSE
Зафиксирована аномалия
Среднеквадратическая ошибка (MSE)

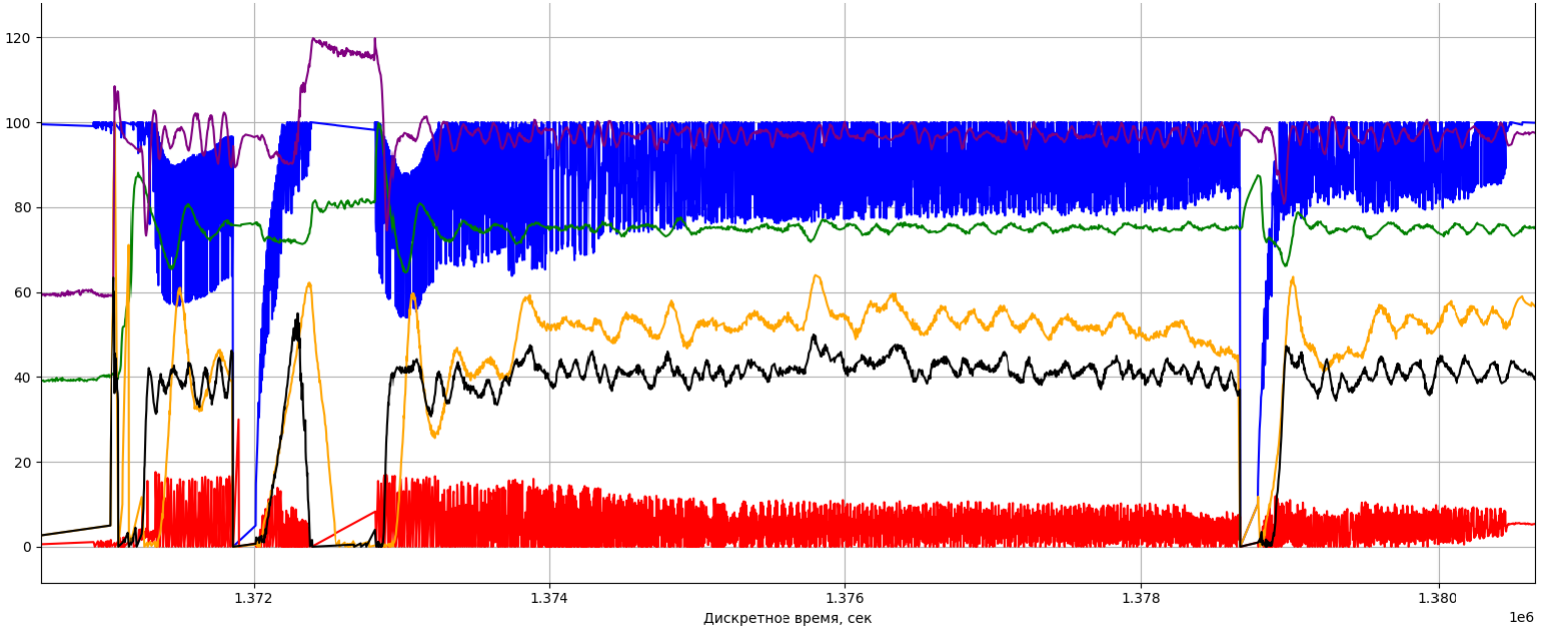
			3
			○
			○


Данные сидов: 1 2 3 4 5 6
Мощность циркуляционного насоса N101, %
Расход FE101, м3/час
Температура гомогенизации TE100, градус Цельсия
Степень открытия парового клапана VC100, %
Температура пастеризации TE101, градус Цельсия
Степень открытия парового клапана VC101, %
120
100
80
60
40
20
0
5.40
5.45
5.50
5.55
5.60
1e6
Дискретное время, сек


Данные сидов: 1 2 3 4 5 6
Мощность циркуляционного насоса N101, %
Расход FE101, м3/час
Температура гомогенизации TE100, градус Цельсия
Степень открытия парового клапана VC100, %
Температура пастеризации TE101, градус Цельсия
Степень открытия парового клапана VC101, %
0
20
40
60
80
100
120
140
1.89
1.90
1.91
1.92
1.93
1.94
1e7
Дискретное время, сек


Данные сидов: 1 2 3 4 5 6
Мощность циркуляционного насоса N101, %
Расход FE101, м3/час
Температура гомогенизации TE100, градус Цельсия
Степень открытия парового клапана VC100, %
Температура пастеризации TE101, градус Цельсия
Степень открытия парового клапана VC101, %
160
140
120
100
80
60
40
20
0
2.5665
2.5670
2.5675
2.5680
2.5685
2.5690
1e7
Дискретное время, сек


125
100
75
50
25
0
1.280
1.282
1.284
1.286
1.288
Дискретное время, сек
1e6


120
100
80
60
40
20
0
1.372
1.374
1.376
1.378
1.380
Дискретное время, сек
1e6

$t_i = t_0 + i \cdot r$

t_0
r
it_i
t_i

$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1}$

$(x, y)(x_1, y_1)(x_2, y_2)$
$x_1 \leq x \leq x_2$

$y = \left(\frac{(x-x_1)\cdot(y_2-y_1)}{x_2-x_1} \right) + y_1$

$y \in [a; b] yz \in [c; d]$

$z = \frac{(y-a)\cdot(b-a)}{d-c} + c$

$y = \frac{(z-c)\cdot(d-c)}{b-a} + a$

$C_t H_t X_t$

$F_t I_t G_t O_t C_t H_t$
$F_t C_t$

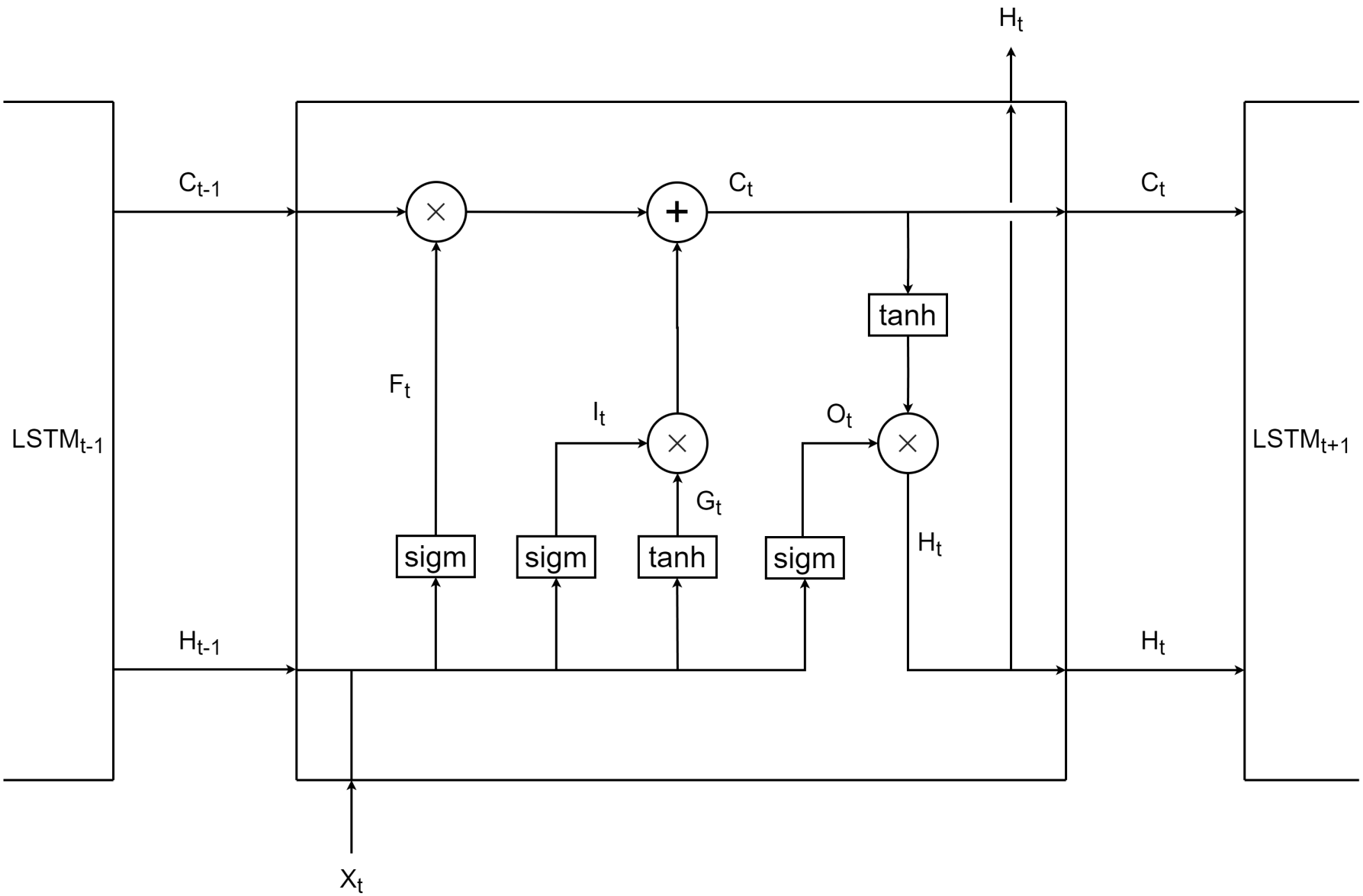
$F_t = sigm(W_f \times X_t + U_f \times H_{t-1} + b_f)$

F_t
X_t
W_f
H_{t-1}
b_f
U_f
$G_t C_t$

$G_t = tanh(W_g \times X_t + U_g \times H_{t-1} + b_g)$

G_t
W_g
b_g
U_g
$I_t G_t C_t$

$I_t = sigm(W_i \times X_t + U_i \times H_{t-1} + b_i)$


H_t
C_{t-1}
C_t
C_t
tanh
F_t
I_t
O_t
$LSTM_{t-1}$
$LSTM_{t+1}$
G_t
sigm
sigm
tanh
sigm
H_t
H_{t-1}
H_t
X_t

I_t
W_i
b_i
U_i

C_t

$$C_t = F_t \times C_{t-1} + I_t \times G_t$$

C_{t-1}

O_t

$$O_t = sigm(W_o \times X_t + U_o \times H_{t-1} + b_o)$$

O_t
W_o
b_o
U_o

$$H_t = O_t \times tanh(C_t)$$

$$Y_t = sigm(W_y \times H_t + b_y)$$

Y_t
W_y
b_y

MSE

$$MSE = 0.5 \cdot (E_t - Y_t)^2$$

Y_t
E_t

$$\frac{\delta MSE}{\delta Y_t} = (Y_t - E_t) \times sigm'(Y_t)$$

$\frac{\delta MSE}{\delta Y_t}$

W_y

$$\frac{\delta MSE}{\delta W_y} = \frac{\delta MSE}{\delta Y_y} \times \frac{\delta Y_t}{\delta W_y} = \frac{\delta MSE}{\delta Y_t} \times H_t$$

$\frac{\delta MSE}{\delta W_y}$

$$\frac{\delta MSE}{\delta b_y} = \frac{\delta MSE}{\delta Y_t} \times \frac{\delta Y_t}{\delta b_y} = \frac{\delta MSE}{\delta Y_t}$$

$$\frac{\delta MSE}{\delta b_y}$$

$$W = W - \alpha \cdot \frac{\delta MSE}{\delta W}$$

$$W$$
$$\frac{\delta MSE}{\delta W}$$

$$b = b + \alpha \cdot \frac{\delta MSE}{\delta b}$$

$$\frac{\delta MSE}{\delta H_t} = \frac{\delta MSE}{\delta Y_t} \times \frac{\delta Y_t}{\delta H_t} + \frac{\delta MSE}{\delta F_{t+1}} \times \frac{\delta F_{t+1}}{\delta H_t} + \frac{\delta MSE}{\delta I_{t+1}} \times \frac{\delta I_{t+1}}{\delta H_t} +$$
$$+ \frac{\delta MSE}{\delta G_{t+1}} \times \frac{\delta G_{t+1}}{\delta H_t} + \frac{\delta MSE}{\delta O_{t+1}} \times \frac{\delta O_{t+1}}{\delta H_t} = \frac{\delta MSE}{\delta Y_t} \times W_y + \frac{\delta MSE}{\delta F_{t+1}} \times U_f +$$
$$+ \frac{\delta MSE}{\delta I_{t+1}} \times U_i + \frac{\delta MSE}{\delta G_{t+1}} \times U_g + \frac{\delta MSE}{\delta O_{t+1}} \times U_o$$

$$\frac{\delta MSE}{\delta H_t}$$
$$\frac{\delta MSE}{\delta F_{t+1}}$$
$$\frac{\delta MSE}{\delta I_{t+1}}$$
$$\frac{\delta MSE}{\delta G_{t+1}}$$
$$\frac{\delta MSE}{\delta O_{t+1}}$$

$$\frac{\delta MSE}{\delta C_t} = \frac{\delta MSE}{\delta H_t} \times \frac{\delta H_t}{\delta C_t} + \frac{\delta MSE}{\delta C_{t+1}} \times \frac{\delta C_{t+1}}{\delta C_t} = \frac{\delta MSE}{\delta H_t} \times$$
$$\times O_t \times tanh'(C_t) + \frac{\delta MSE}{\delta C_{t+1}} \times F_{t+1}$$

$$\frac{\delta MSE}{\delta C_t}$$
$$\frac{\delta MSE}{\delta C_{t+1}}$$
$$F_{t+1}$$

$$\frac{\delta MSE}{\delta F_t} = \frac{\delta MSE}{\delta C_t} \times \frac{\delta C_t}{\delta F_t} = \frac{\delta MSE}{\delta C_t} \times C_{t-1} \times sigm'(F_t)$$

$$\frac{\delta MSE}{\delta F_t}$$

$$\frac{\delta MSE}{\delta W_f} = \frac{\delta MSE}{\delta F_t} \times \frac{\delta F_t}{\delta W_f} = \frac{\delta MSE}{\delta F_t} \times X_t$$

$$\frac{\delta MSE}{\delta W_f}$$

$$\frac{\delta MSE}{\delta U_f} = \frac{\delta MSE}{\delta F_t} \times \frac{\delta F_t}{\delta U_f} = \frac{\delta MSE}{\delta F_t} \times H_{t-1}$$

$$\frac{\delta MSE}{\delta U_f}$$

$$\frac{\delta MSE}{\delta b_f} = \frac{\delta MSE}{\delta F_t} \times \frac{\delta F_t}{\delta b_f} = \frac{\delta MSE}{\delta F_t}$$

$$\frac{\delta MSE}{\delta I_t} = \frac{\delta MSE}{\delta C_t} \times \frac{\delta C_t}{\delta I_t} = \frac{\delta MSE}{\delta C_t} \times G_t \times sigm'(I_t)$$

$$\frac{\delta MSE}{\delta I_t}$$

$$\frac{\delta MSE}{\delta W_i} = \frac{\delta MSE}{\delta I_t} \times \frac{\delta I_t}{\delta W_i} = \frac{\delta MSE}{\delta I_t} \times X_t$$

$$\frac{\delta MSE}{\delta W_i}$$

$$\frac{\delta MSE}{\delta U_i} = \frac{\delta MSE}{\delta I_t} \times \frac{\delta I_t}{\delta U_i} = \frac{\delta MSE}{\delta I_t} \times H_{t-1}$$

$$\frac{\delta MSE}{\delta U_i}$$

$$\frac{\delta MSE}{\delta b_i} = \frac{\delta MSE}{\delta I_t} \times \frac{\delta I_t}{\delta b_i} = \frac{\delta MSE}{\delta I_t}$$

$$\frac{\delta MSE}{\delta G_t} = \frac{\delta MSE}{\delta C_t} \times \frac{\delta C_t}{\delta G_t} = \frac{\delta MSE}{\delta C_t} \times I_t \times tanh'(G_t)$$

$$\frac{\delta MSE}{\delta G_t}$$

$$\frac{\delta MSE}{\delta W_g} = \frac{\delta MSE}{\delta G_t} \times \frac{\delta G_t}{\delta W_g} = \frac{\delta MSE}{\delta G_t} \times X_t$$

$$\frac{\delta MSE}{\delta W_g}$$

$$\frac{\delta MSE}{\delta U_g} = \frac{\delta MSE}{\delta G_t} \times \frac{\delta G_t}{\delta U_g} = \frac{\delta MSE}{\delta G_t} \times H_{t-1}$$

$$\frac{\delta MSE}{\delta U_g}$$

$$\frac{\delta MSE}{\delta b_g} = \frac{\delta MSE}{\delta G_t} \times \frac{\delta G_t}{\delta b_g} = \frac{\delta MSE}{\delta G_t}$$

$$\frac{\delta MSE}{\delta O_t} = \frac{\delta MSE}{\delta H_t} \times \frac{\delta H_t}{\delta O_t} = \frac{\delta MSE}{\delta H_t} \times tanh(C_{t-1}) \times sigm'(O_t)$$

$$\frac{\delta MSE}{\delta O_t}$$

$$\frac{\delta MSE}{\delta W_o} = \frac{\delta MSE}{\delta O_t} \times \frac{\delta O_t}{\delta W_o} = \frac{\delta MSE}{\delta O_t} \times X_t$$

$$\frac{\delta MSE}{\delta W_o}$$

$$\frac{\delta MSE}{\delta U_o} = \frac{\delta MSE}{\delta O_t} \times \frac{\delta O_t}{\delta U_o} = \frac{\delta MSE}{\delta O_t} \times H_{t-1}$$

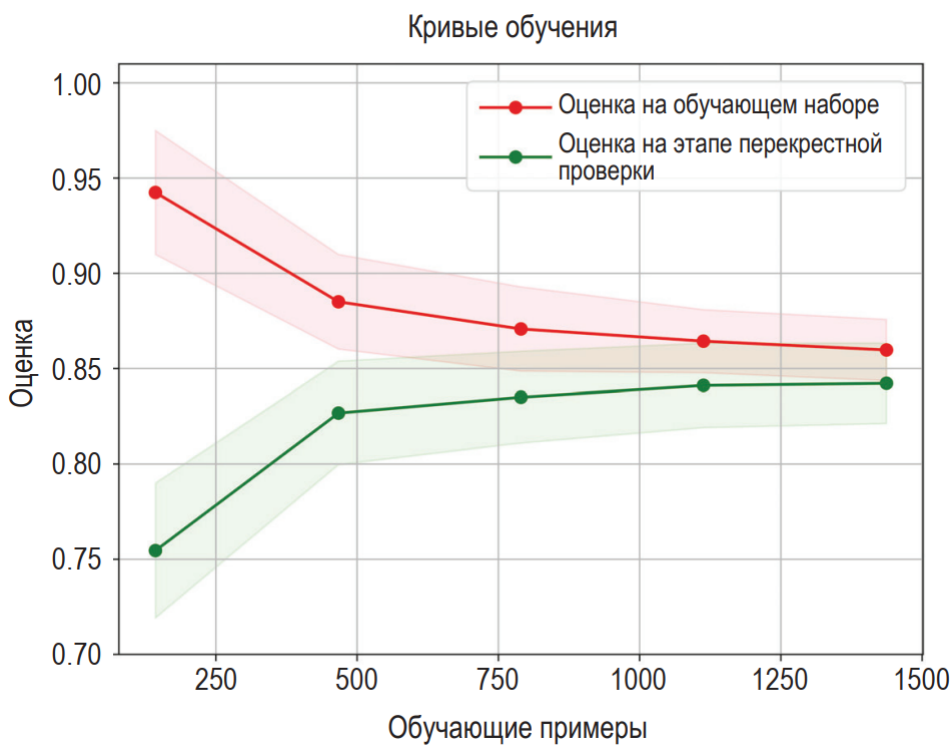
$$\frac{\delta MSE}{\delta U_o}$$

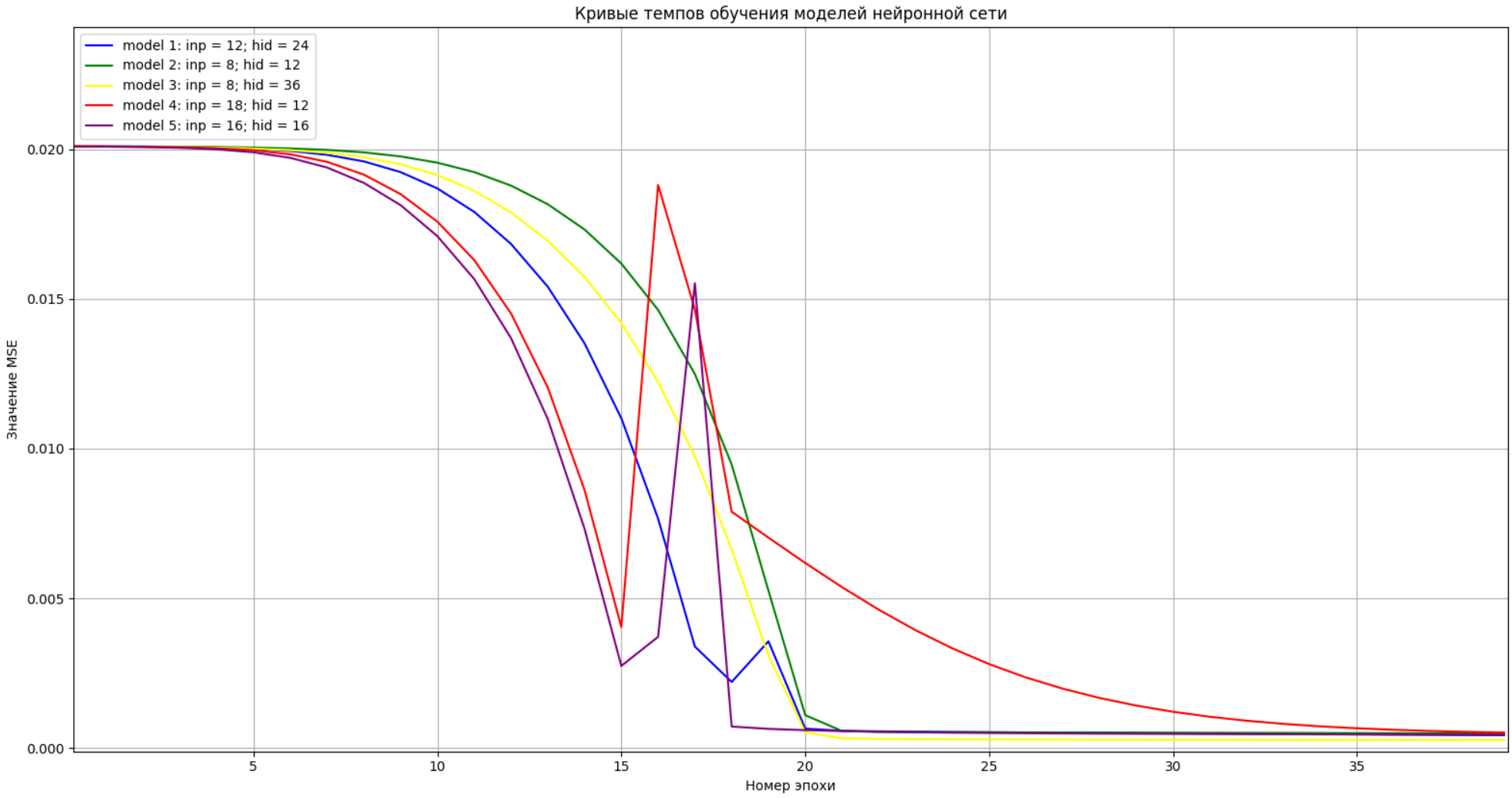
$$\frac{\delta MSE}{\delta b_o} = \frac{\delta MSE}{\delta O_t} \times \frac{\delta O_t}{\delta b_o} = \frac{\delta MSE}{\delta O_t}$$

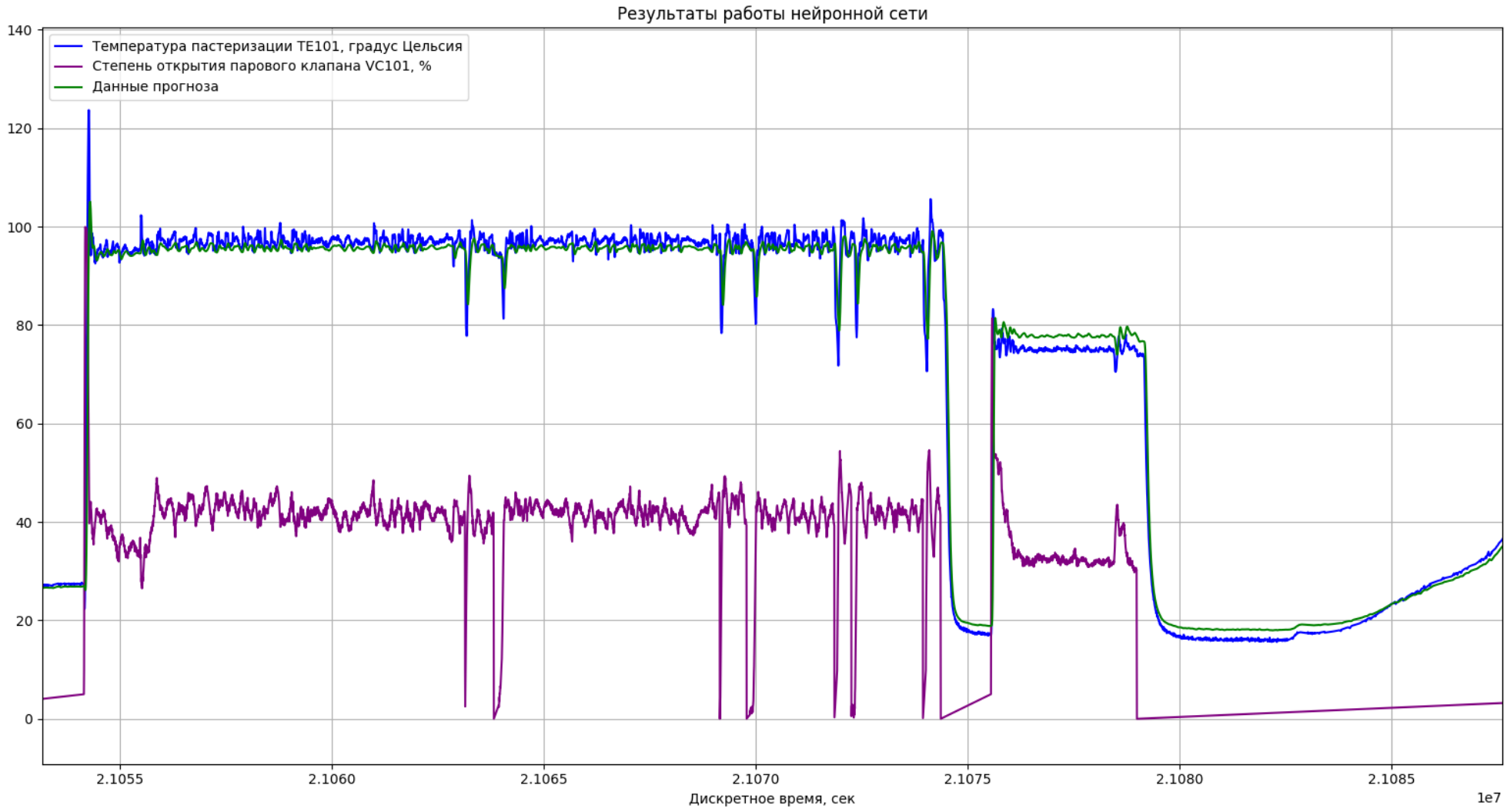
$inpX_t$
$hidF_tO_tI_tG_tC_tH_t$
$outY_t$
$[hid \times 1][hid \times inp]$
α
TE
$epsTE$
TS

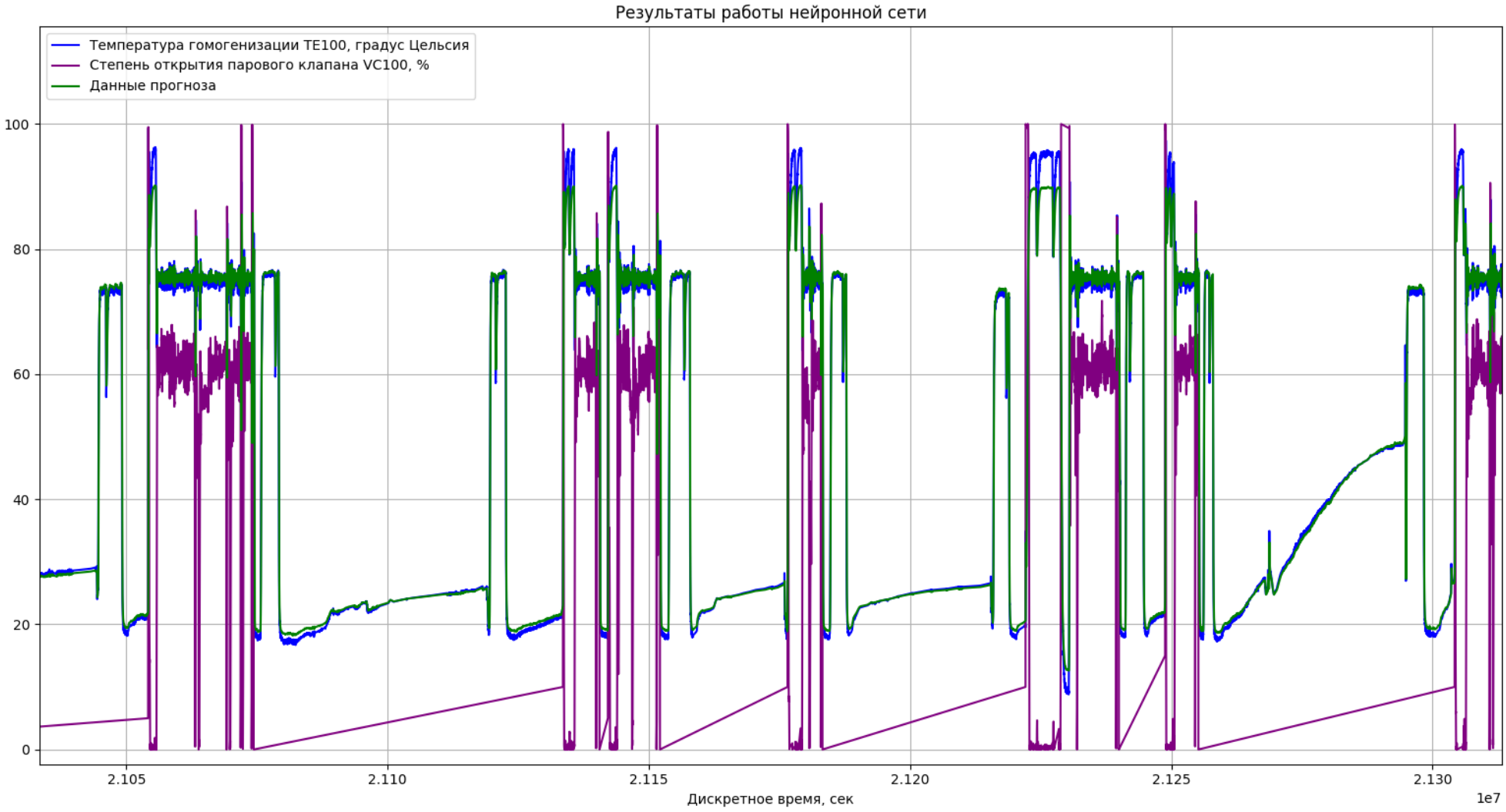
```
D:\Prog\PAIA\NNTrainer\out\build\x64-Debug\NNTrainer.exe
epoch = 1462/1500 | time = 4.011 sec | E = 6.80222e-05
epoch = 1463/1500 | time = 4 sec | E = 6.7873e-05
epoch = 1464/1500 | time = 4.031 sec | E = 6.77247e-05
epoch = 1465/1500 | time = 4.07 sec | E = 6.75775e-05
epoch = 1466/1500 | time = 4.044 sec | E = 6.74312e-05
epoch = 1467/1500 | time = 4.093 sec | E = 6.72858e-05
epoch = 1468/1500 | time = 4.022 sec | E = 6.71415e-05
epoch = 1469/1500 | time = 4.009 sec | E = 6.69981e-05
epoch = 1470/1500 | time = 4.074 sec | E = 6.68556e-05
epoch = 1471/1500 | time = 3.989 sec | E = 6.67141e-05
epoch = 1472/1500 | time = 4.017 sec | E = 6.65735e-05
epoch = 1473/1500 | time = 3.992 sec | E = 6.64338e-05
epoch = 1474/1500 | time = 4 sec | E = 6.6295e-05
epoch = 1475/1500 | time = 4.001 sec | E = 6.61572e-05
epoch = 1476/1500
```

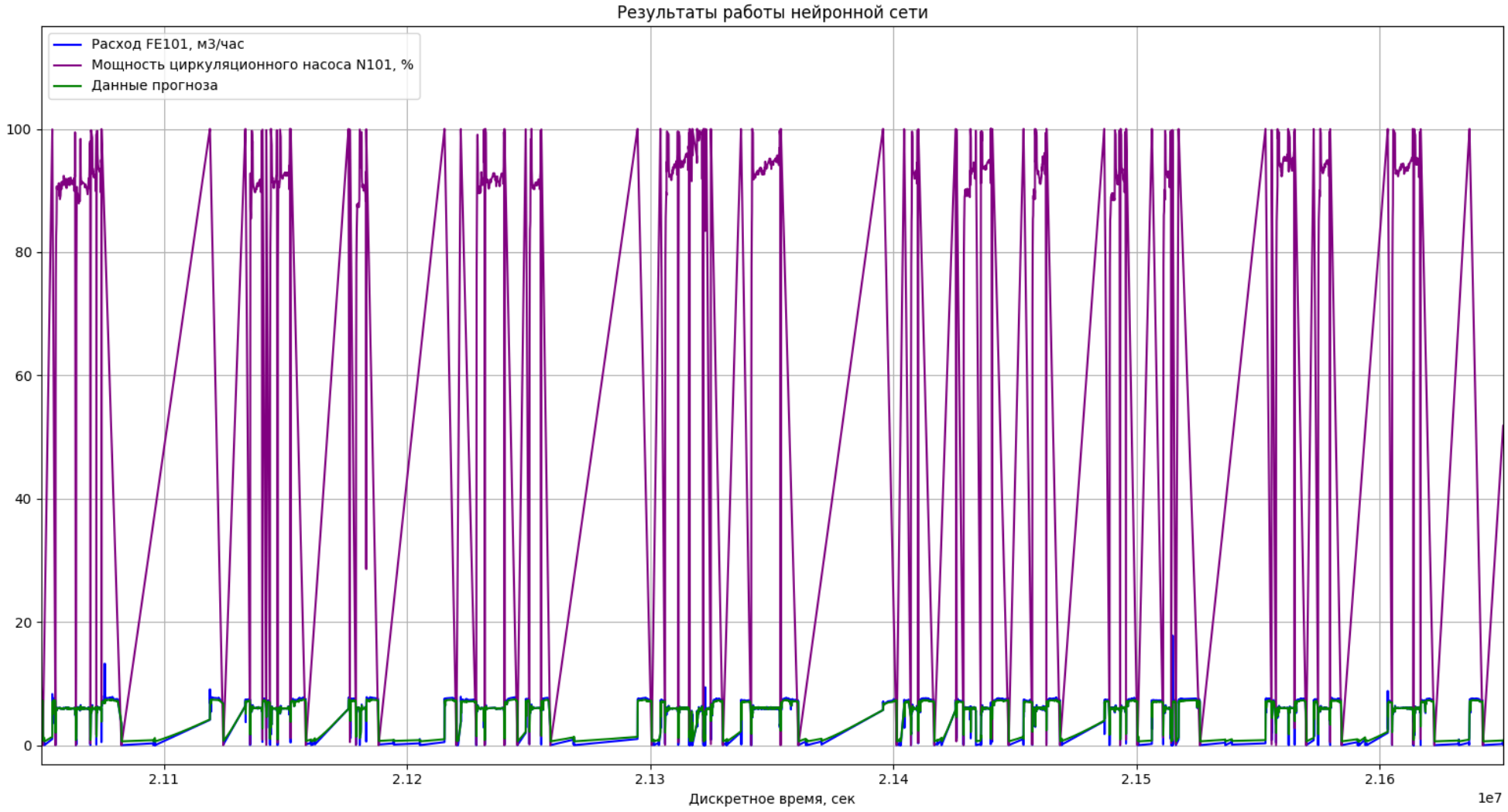
$\alpha = 0.0001 TE = 0.00001 eps = 40 out = 4 TS = 200000$



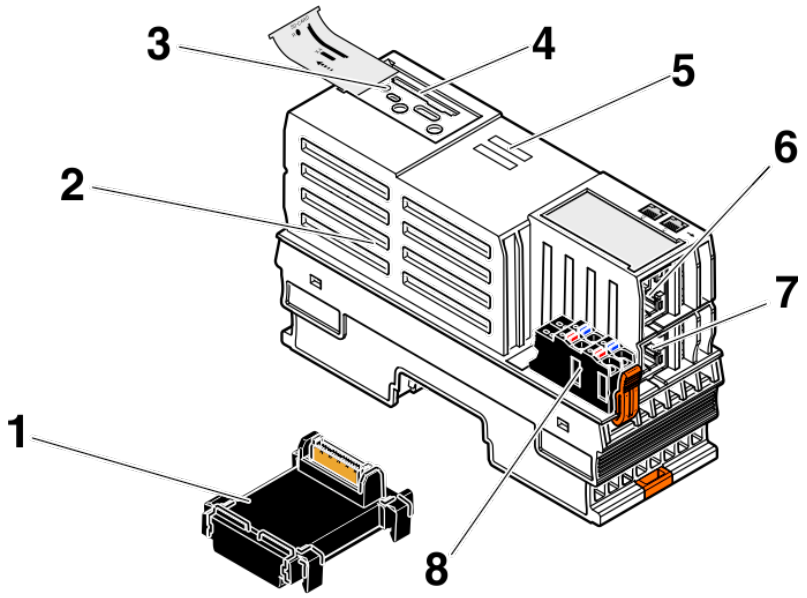

Кривые темпов обучения моделей нейронной сети
model 1: inp = 12; hid = 24
model 2: inp = 8; hid = 12
model 3: inp = 8; hid = 36
model 4: inp = 18; hid = 12
model 5: inp = 16; hid = 16
Значение MSE
0.020
0.015
0.010
0.005
0.000
5
10
15
20
25
30
35
Номер эпохи


Результаты работы нейронной сети
Температура пастеризации TE101, градус Цельсия
Степень открытия парового клапана VC101, %
Данные прогноза
140
120
100
80
60
40
20
0
2.1055
2.1060
2.1065
2.1070
2.1075
2.1080
2.1085
1e7
Дискретное время, сек


Результаты работы нейронной сети
Температура гомогенизации TE100, градус Цельсия
Степень открытия парового клапана VC100, %
Данные прогноза
100
80
60
40
20
0
2.105
2.110
2.115
2.120
2.125
2.130
1e7
Дискретное время, сек


Результаты работы нейронной сети
Расход FE101, м3/час
Мощность циркуляционного насоса N101, %
Данные прогноза
100
80
60
40
20
0
2.11
2.12
2.13
2.14
2.15
2.16
1e7
Дискретное время, сек

PLCnext Technology
Designed by PHOENIX CONTACT
Ecosystem for limitless automation
10V
0V
GETTING STARTED
starterkit.plcnextcommunity.com


3
4
5
6
2
7
1
8

192.168.1.10 - PuTTY

```
admin@axcf2152:~$ ./ForecasterTS
time = 1


time = 2


time = 3


time = 4
```

192.168.1.10 - PuTTY

```
predicts:       94.5891 94.5379 94.4828 94.4754 94.4125

time = 84

taeget:         95.9667 96.1333 96.3000 96.1667 96.2333 96.1000 96.2333 96.3000 97.6000 98.8667 99.2333 98.9000 98.2333 97.7667 97.5000 97.2500
other:          43.2500 44.0000 44.1000 43.8600 43.6200 44.2000 44.8000 43.9000 43.1000 42.8000 42.4000 42.0000 41.6000 41.9000 41.7000 42.0333
predicts:       94.6349 94.5837 94.5286 94.5211 94.4582

time = 85

taeget:         96.1333 96.3000 96.1667 96.2333 96.1000 96.2333 96.3000 97.6000 98.8667 99.2333 98.9000 98.2333 97.7667 97.5000 97.2500 97.0500
other:          44.0000 44.1000 43.8600 43.6200 44.2000 44.8000 43.9000 43.1000 42.8000 42.4000 42.0000 41.6000 41.9000 41.7000 42.0333 42.9000
predicts:       94.6721 94.6209 94.5657 94.5583 94.4954

time = 86

taeget:         96.3000 96.1667 96.2333 96.1000 96.2333 96.3000 97.6000 98.8667 99.2333 98.9000 98.2333 97.7667 97.5000 97.2500 97.0500 96.9000
other:          44.1000 43.8600 43.6200 44.2000 44.8000 43.9000 43.1000 42.8000 42.4000 42.0000 41.6000 41.9000 41.7000 42.0333 42.9000 42.7923
predicts:       94.7006 94.6494 94.5943 94.5868 94.5239
```

$V_0 = \sum_{i=0}^{n} V_i$

V_0

V_i

$V_y = \sum_{i=0}^{n} V_{yi}$

V_y

V_{yi}

		V_y	V_{yi}

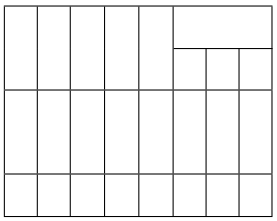
V_{yi}

$= \cdot \cdot \cdot \cdot$

$\cdot \cdot \cdot$

$= \cdot \cdot$

$= \cdot$



$= (+) \cdot$

$= \cdot$

$= \cdot \cdot$

$= \cdot$

$= \cdot \cdot \cdot$

$= \cdot \cdot$

$= \cdot \cdot$

$= \cdot$

$+ + + + + + + +$

$= \cdot \cdot$

$= \cdot$

$= \cdot \cdot$

$= \cdot$

$= + +$

$= \cdot$

$\cdot$

$= +$

$=$

$= + +$

$= +$

$= \ +$

$=$

$= \cdot (- \cdot)$

$= \cdot (- \ )$